AMAS Exam Solutions

Sagar Lee Stockwell Clemensen (FCK916)

Advanced Methods in Applied Statistics

March 26-27, 2026

I, Sagar Lee Stockwell Clemensen, expressly vow to uphold my scientific and academic integrity by working individually on this exam and soliciting no direct external help or assistance

AI Declaration: I used Claude.ai to help format tables for the latex file.

Problem 1

Problem 1a

Using the data from the first, third, and fourth columns, I have identified the function from which each was generated. The data from column 1 was fitted with the function

$$f(x) = sin(ax + d) + cexp(bx) + 1, \tag{1}$$

the data from column 3 with

$$\frac{\lambda^k e^{-\lambda}}{k!}, \tag{2}$$

and the data from column 4 with

$$exp\left(\frac{-(x-a)^2}{2b^2}\right). \tag{3}$$

The best-fit values of the parameters and their 1σ confidence uncertainties are presented in Tab. 1. The uncertainties are obtained via bootstrapping. A p-value is also presented. This is computed by sampling from the fitted PDF and using a Kolmogorov-Smirnov two-sample test on the sampled distribution and real data. Since all p-values are above 0.25, the identified functions cannot be rejected as good fits to the data in the respective column at 2σ confidence.

	Column 1 Eq. 1	**Column 3** Eq. 2	**Column 4** Eq. 3
a	$5.29^{+0.04}_{-0.06}$	-	1.1385 ± 0.0005
b	$-0.99^{+0.03}_{-0.04}$	-	$0.0402^{+0.04}_{-0.05}$
c	5184^{+1222}_{-920}	-	-
d	$0.1^{+0.6}_{-0.4}$	-	-
λ	-	$6.12^{+0.04}_{-0.03}$	-
p-value	0.2559	0.8982	0.8255

Table 1: Values of fitted parameters with uncertainties from bootstrapping. By sampling from the fitted PDF and comparing this sampled distribution with the real data a Kolmogorov-Smirnov two-sample test is used to compute a p-value.

Problem 1b

Figures 1, 2, and 3 show the data from columns 1, 3, and 4 alongside their respective fitted PDFs. Each legend includes the fitted PDF expression, the $-LLH$ value, and parameter uncertainties from the fit assuming Gaussian errors, rather than those from bootstrapping. The two sets of uncertainties are in close agreement, and using the Gaussian estimates keeps the legends uncluttered.

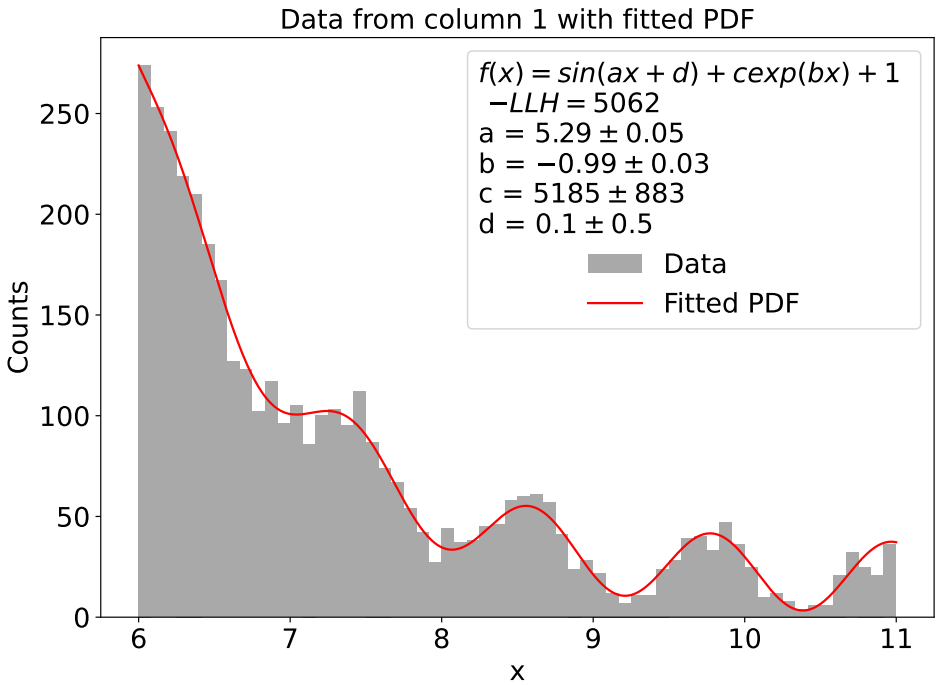

Figure 1: Data from column 1 with fitted PDF. The fitted PDF, $-LLH$ value for the fit, and parameter values with uncertainties are written in the legend.

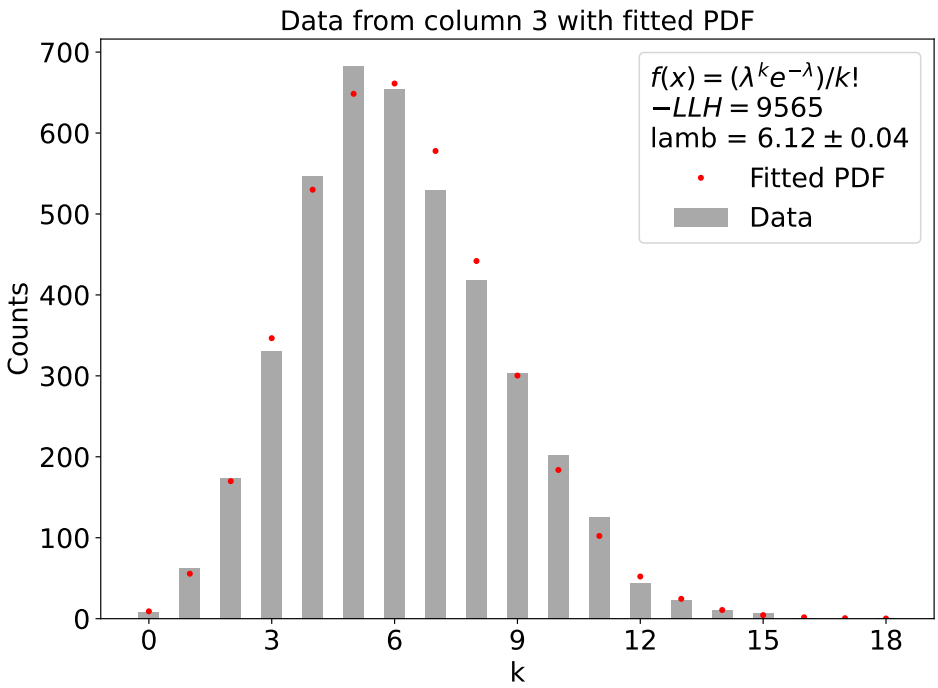

Figure 2: Data from column 3 with fitted PDF. The fitted PDF, $-LLH$ value for the fit, and parameter values with uncertainties are written in the legend.

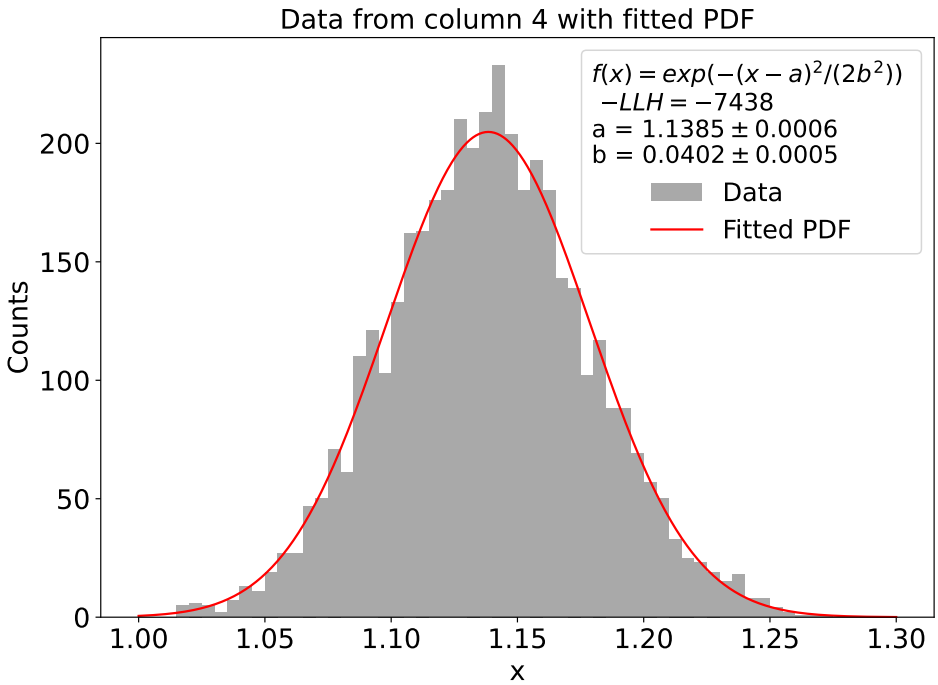

Figure 3: Data from column 4 with fitted PDF. The fitted PDF, $-LLH$ value for the fit, and parameter values with uncertainties are written in the legend.

Problem 2

Problem 2a

To determine whether the data is spherically isotropically distributed, I compute a Kolmogorov-Smirnov (KS) statistic for the real data by applying a KS test to the cumulative two-point autocorrelation function, comparing it against that expected for an isotropic distribution. I then generate 2000 isotropically sampled datasets, compute the KS statistic for each, and obtain a reference distribution of KS statistics that follow the isotropic hypothesis. The p-value is then the fraction of simulated KS-statistics exceeding that of the KS statistic of the real data. The resulting p-value of 0.0025 is sufficient to reject the isotropic hypothesis at 3σ confidence. The distribution of KS statistics is presented in Fig. 4.

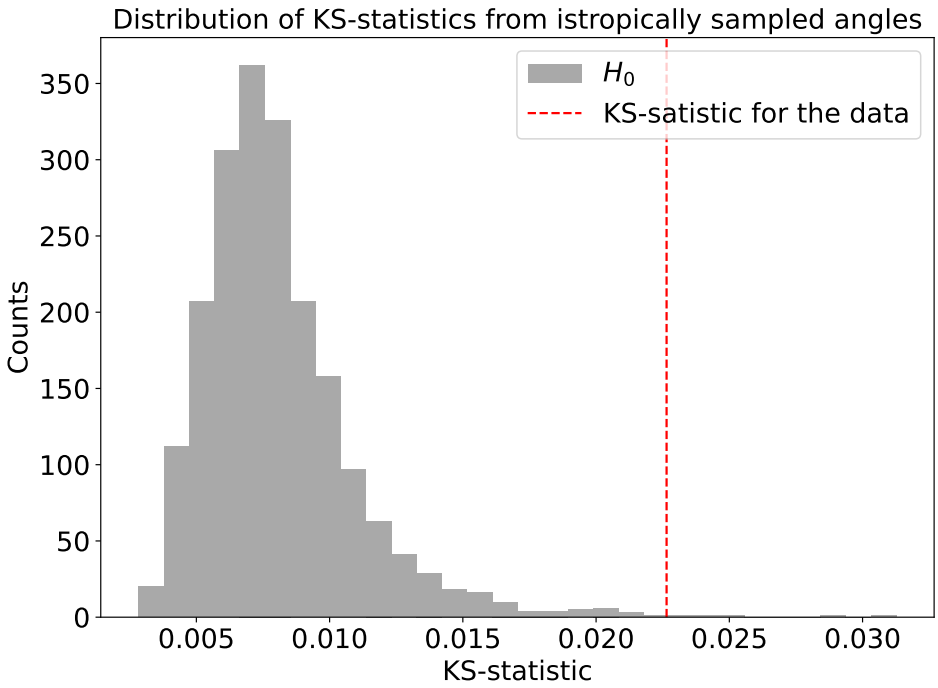

Figure 4: Distribution of KS statistics of sampled spherically isotropically distributed azimuthal and zenith angles. The KS statistic is calculated using the cumulative two-point autocorrelation function.

Problem 2b

To test whether the data fits the H_a and H_b hypotheses better than the isotropic hypothesis H_0 I once again generate data according to each of the hypotheses and compute a KS statistic for 2000 trials. The distribution of KS statistics for H_0, H_a, and H_b are presented in Fig. 5. Additionally I have computed the p-values of H_0 vs H_a and H_b as well as the p-values for whether the real data fits either H_a or H_b. The results are presented in Tab. 2. As a cross-check, the p-values of H_0 vs H_a and H_b confirm that both alternative hypotheses are inconsistent with isotropy. By construction both H_a and H_b were defined as anisotropic distributions, so this was expected.

With a p-value of 0.8360 for H_a the hypothesis that the data is distributed according to H_a cannot be rejected even at 1σ confidence. For H_b the p-value is a bit lower at 0.2250 still the hypothesis that the data is distributed according to H_b can only be rejected at 1.2σ confidence. Both hypotheses fit the data much better than the isotropic hypothesis and neither can be rejected at 2σ confidence.

	H_a	H_b
H_0	0.0005	0.0190
Real data	0.8360	0.2250

Table 2: p-values of H_0 vs H_a and H_b as well as the p-values for whether the real data fits either H_a or H_b.

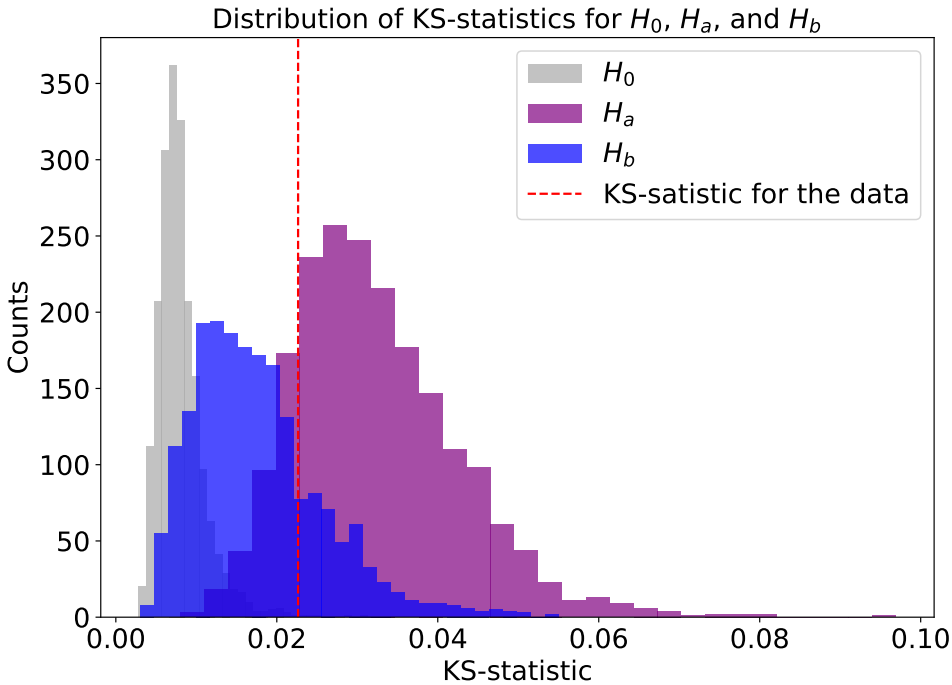

Figure 5: Distribution of KS-statistics for H_0, H_a, and H_b

Problem 3

Problem 3a

A plot of the population estimate as a function of distance along SH1 using KDE with Gaussian kernels that have a bandwidth of 25 km is presented in Fig. 6.

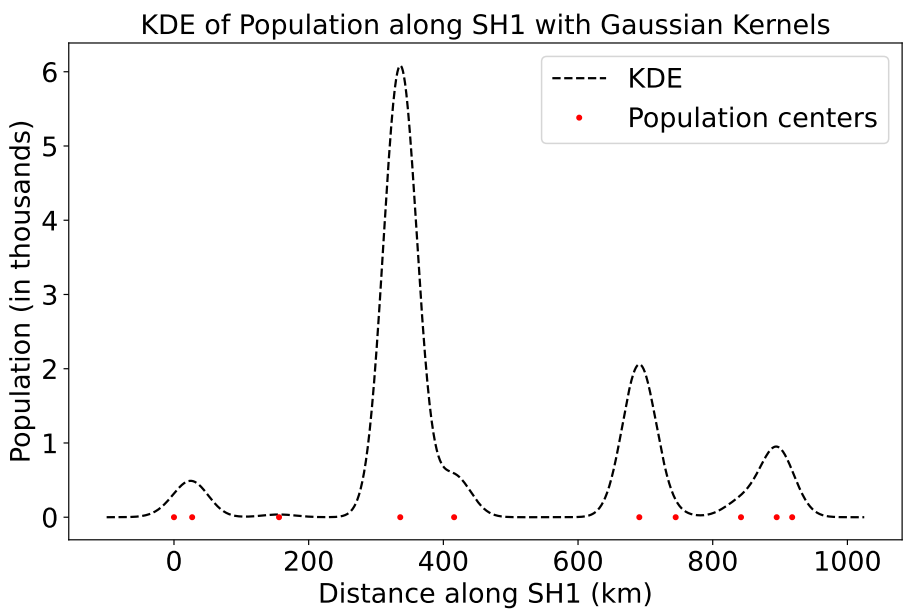

Figure 6: KDE of Population along SH1 with Gaussian kernels that have a bandwidth of 25 km.

Using the KDE, the total population estimate from 405 km to 430 km along SH1 is 14400.

Problem 3b

The computed amount of e-cars in each of the 4 regions along SH1 is presented in Tab. 3.

Region	Amount of e-cars
Nelson-Marlborough	144
Canterbury	2426
Otago	803
Southland	131

Table 3: Amount of e-cars in each of the 4 regions along SH1. Computed using the population KDE shown in Fig. 6.

Problem 3c

A plot of the number of e-cars as a function of distance along SH1 using KDE with Epanechnikov kernels that have a bandwidth of 35 km is is presented in Fig. 7.

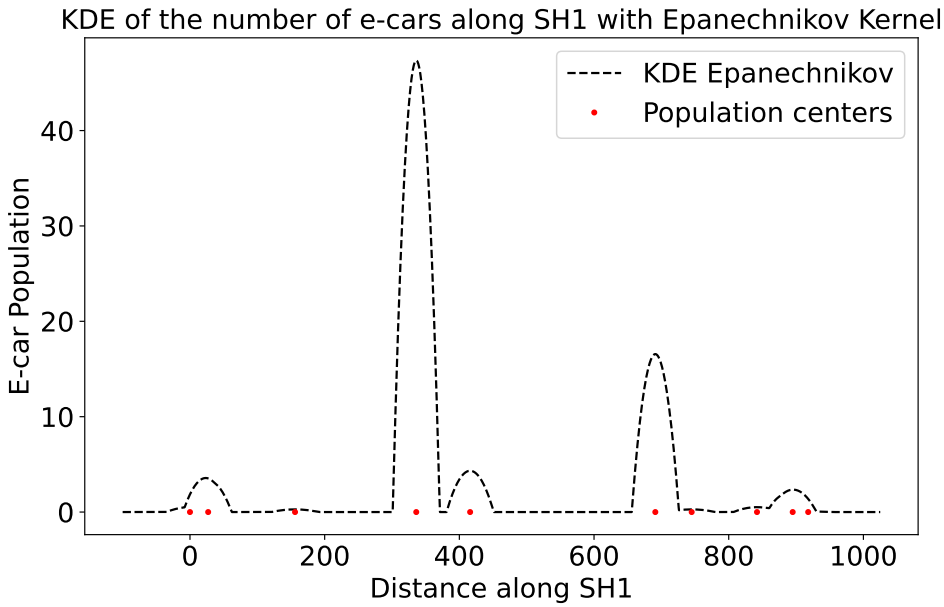

Figure 7: KDE of the number of e-cars along SH1 with Epanechnikov kernels that have a bandwidth of 35 km.

A plot of the posterior distribution, where the prior comes from the KDE shown in Fig. 7 and the likelihood comes from a KDE with Gaussian kernels using the camera data, is presented in Fig. 8.

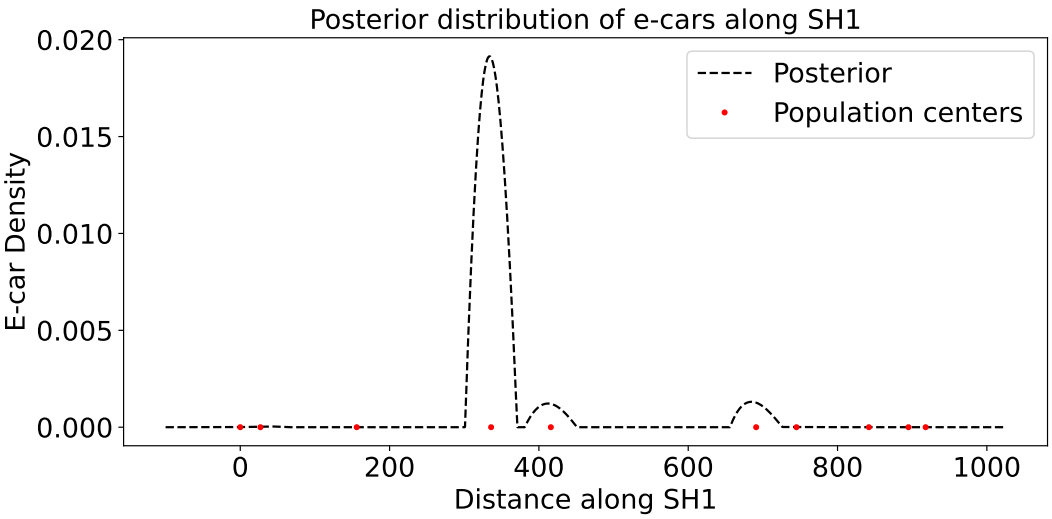

Figure 8: Posterior distribution of e-cars along SH1. The prior comes from the KDE shown in Fig. 7 and the likelihood comes from a KDE with Gaussian kernels using the camera data.

Using the posterior, the fraction of e-cars in New Zealand along SH1 between 400 km to 700 km is 0.089.

A plot showing the posterior, prior, and likelihood is presented in Fig. 9.

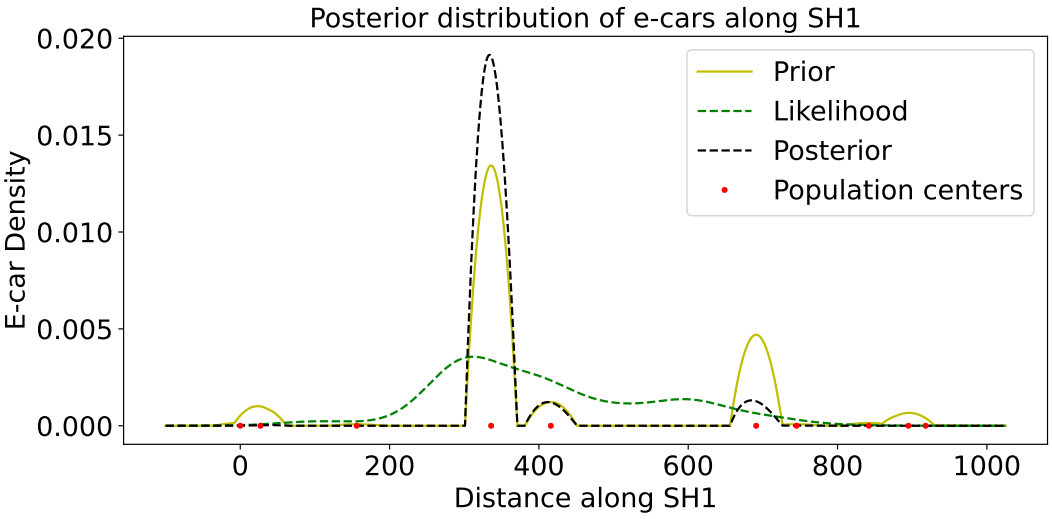

Figure 9: Posterior, prior and likelihood of e-cars along SH1.

Problem 4

Problem 4a

Hypothesis H_a was fitted with a double Gaussian with weights w and $1 - w$, giving 5 free parameters. Hypothesis H_b was fitted with a triple Gaussian with weights w_1, w_2, and $1 - w_1 - w_2$, giving 8 free parameters; 3 more than H_a.

A plot of the glacier elevation data along the fitted PDFs for the hypotheses H_a and H_b is presented in Fig. 10. The $-LLH$ of each fit is listed in the legend.

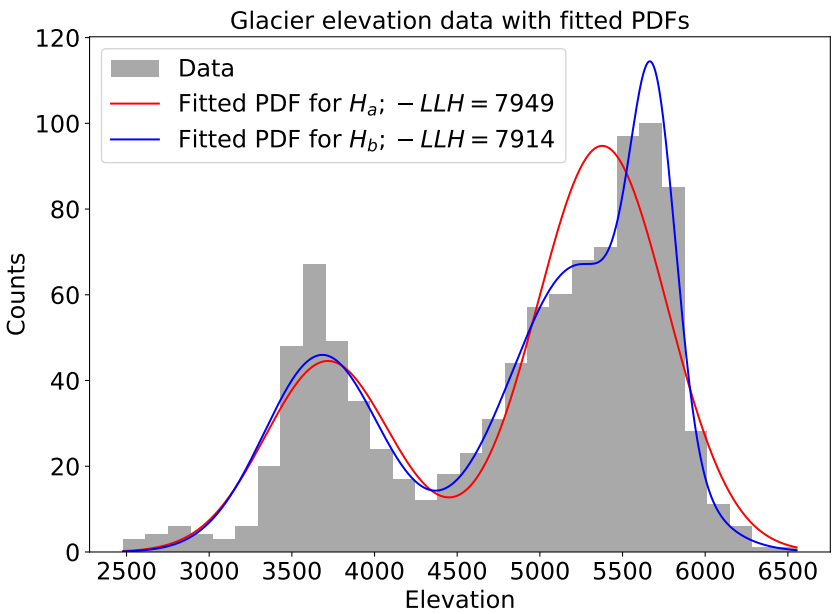

Figure 10: The fitted PDF for hypothesis H_a and H_b alongside the glacier elevation data. The $-LLH$ of each fit is listed in the legend.

Problem 4b

The likelihood ratio statistic is $-2\ln(\lambda) = 71$. Since H_b has 3 more free parameters than H_a, the test statistic follows a χ^2 distribution with 3 degrees of freedom under H_a. The probability of obtaining a value of 71 or higher yields $p = 2.23 \times 10^{-15}$, and H_a can therefore be rejected at 7.9σ confidence.

Problem 5

Problem 5a

For this problem I created a linear and a cubic spline. In Tab. 4 the estimates of the temperature on sol 203.570 is shown. The estimates are from the created linear spline and cubic spline.

Spline type	Temperature (°C)
Linear	-115.913 ± 0.001
Cubic	-115.325 ± 0.001

Table 4: Estimates of the temperature on sol 203.570 from the created linear spline and cubic spline.

Problem 5b

A scatter plot of temperature versus time is presented in Fig. 11, displaying interpolated values from both the linear and cubic splines covering the time range from 203.410 sol to 203.608 sol.

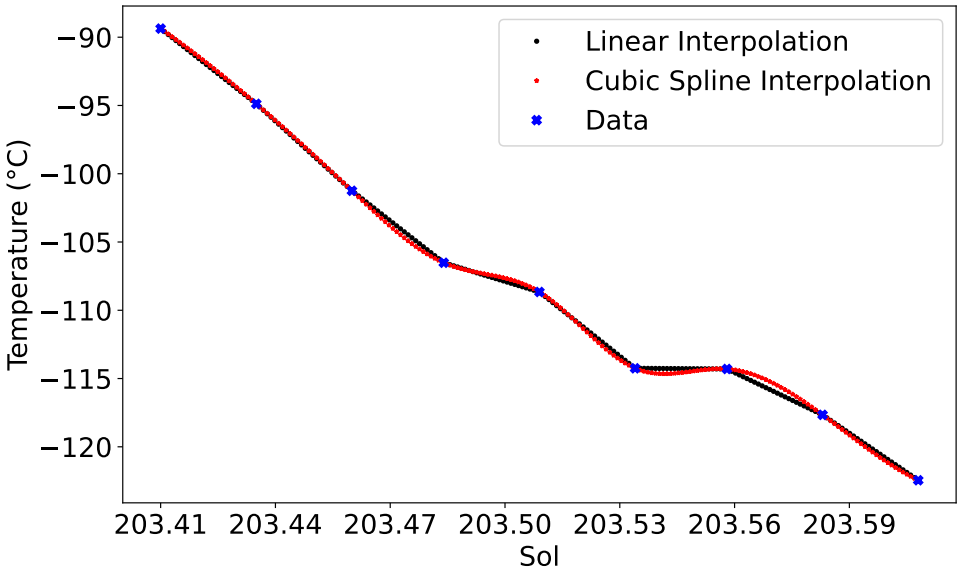

Figure 11: A scatter plot of temperature versus time displaying interpolated values from both the linear and cubic splines.

If we know that during sol 203.410 to 203.608, the temperature should be continuously dropping, we should more closely at the cubic spline in the interval of sol 203.542 to 203.556. Here the interpolated temperature is monotonically increasing, which it shouldn't. This is due to over correction of the cubic spline. A plot of the interpolated values of temperature versus time from the cubic spline, with the monotonically increasing segment highlighted, is presented in Fig. 12.

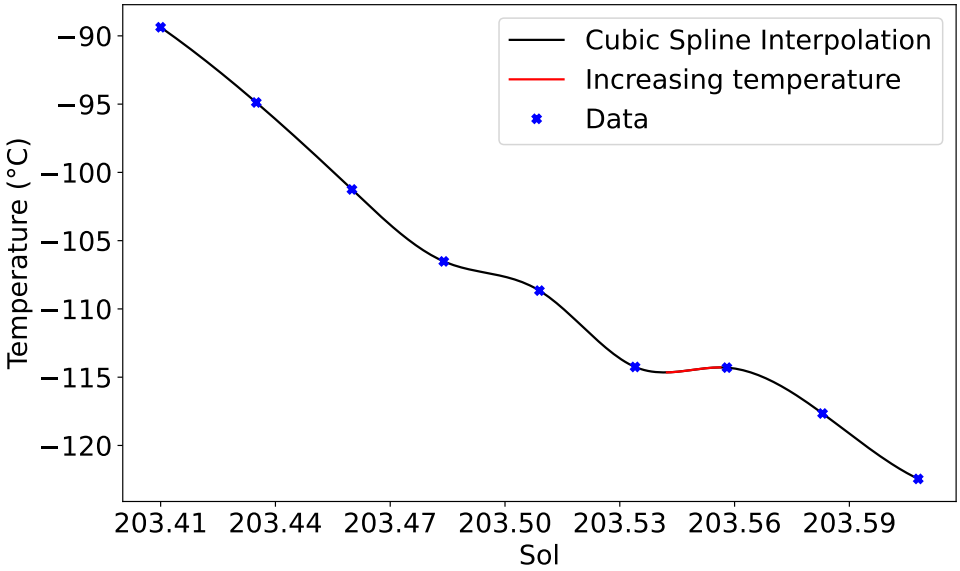

Figure 12: A plot of temperature versus time displaying interpolated values from the cubic spline. The monotonically increasing segment is highlighted with red.

A PCHIP spline is monotonically changing so it doesn't have the same issue as the cubic spline. A plot of one is presented in Fig. 13.

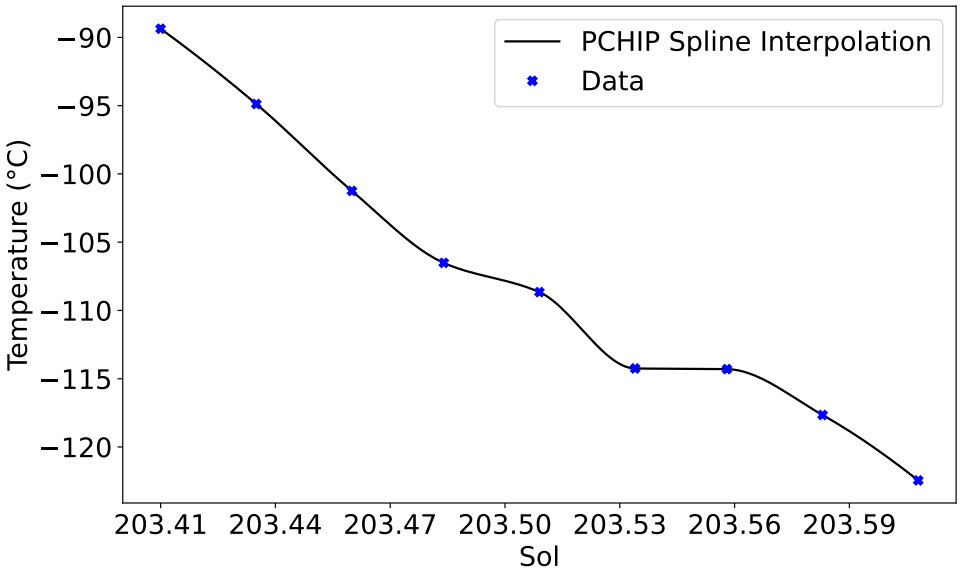

Figure 13: A plot of temperature versus time displaying interpolated values from the PCHIP spline. The monotonically increasing segment is highlighted with red.

Problem 5c

The maximum absolute temperate change within 0.0004 sol for each spline is presented in Tab. 5. For both splines I get an absolute temperature change above 0.1 $^{\circ}C$. Since the proposed electronics components for a Mars rover are unable sustain temperature changes above 0.09 $^{\circ}C$, new electronics with more robustness to temperature fluctuations would be needed according to my interpolations to the Viking 1 data.

Spline type	Greatest absolute temperature change (°C)
Linear	0.102
Cubic	0.106

Table 5: The biggest absolute temperature change within 0.0004 sol from the created linear spline and cubic spline.